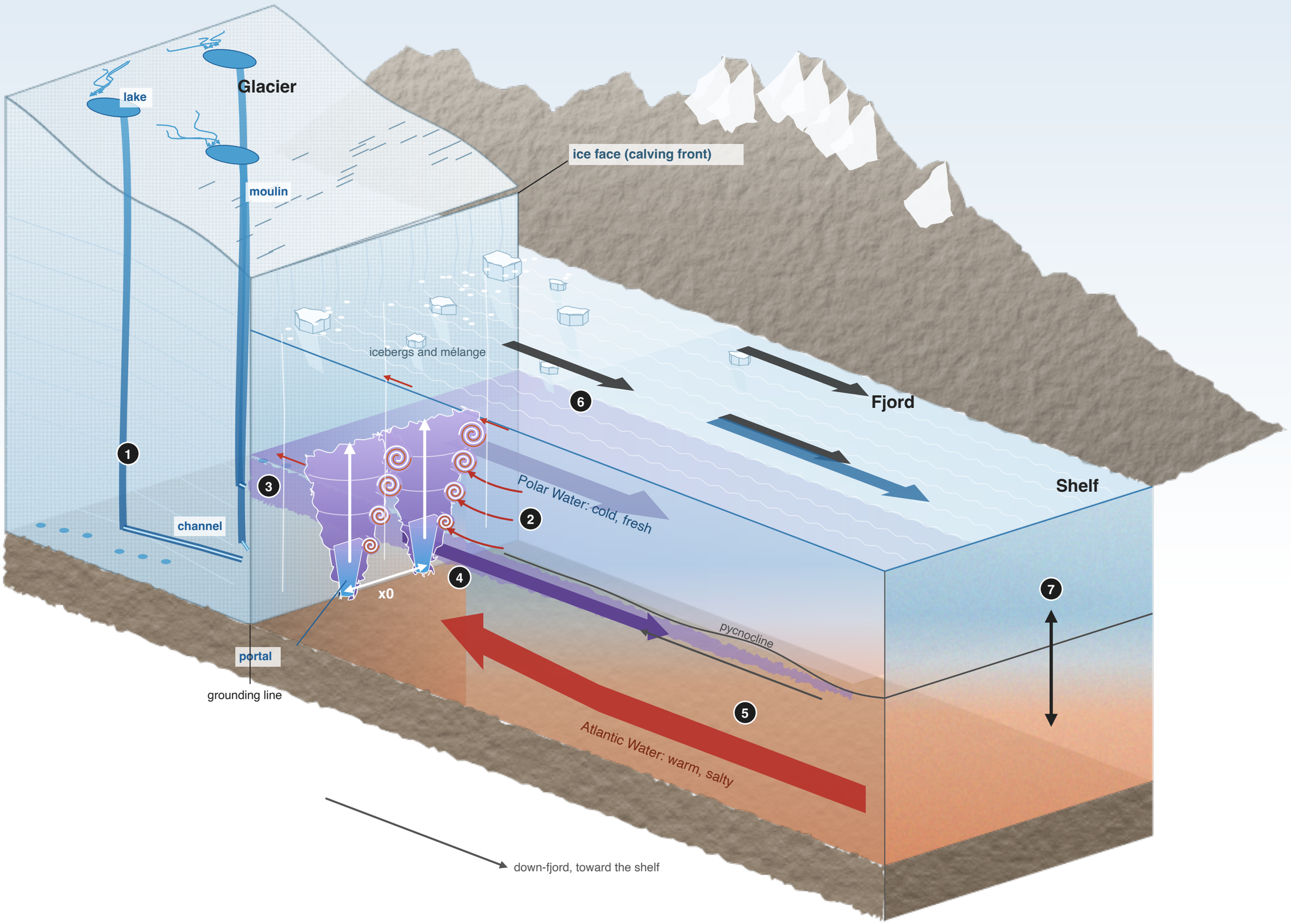
# Processes at a tidewater glacier and in its fjord (3-D view)

Schematic, not to scale: vertical exaggeration; plumes and icebergs enlarged. The near fjord wall is cut away and the water drawn translucent; channels and moulins are shown through the ice.




1. subglacial discharge: lakes drain via moulins to channels at the bed and leave through portals
2. buoyant plumes: rise along the ice; edge eddies engulf ambient water (entrainment)
3. submarine melting: the plumes, and a thin meltwater layer, carry heat to the ice face
4. neutral buoyancy: the plumes merge and spread down-fjord as an outflow layer
5. warm Atlantic Water flows toward the glacier at depth, replacing the entrained water
6. along-fjord wind drives a surface outflow and an exchange with the shelf
7. shelf forcing: the pycnocline heaves over days and the motion enters the fjord